

- Use this 1000, 2020, 1500, 4500, 1050 as array
- Find 4500.
- Use Conditional Jumps and Compare
- Use Compare statement for both termination of loop and compare numbers.
- Use of CX is Prohibited
- Find element and write memory location of value in memory.

The value is stored at: 0109

AX	1194	SI	0109	CS	19F5	IP	012C	Stack	+0 0000	Flags	7204																												
BX	0109	DI	0000	DS	19F5				+2 20CD																														
CX	0031	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF	DF	IF	SF	ZF	AF	PF	CF																						
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0	0	1	0	0	0	1	0																						
S or SI or SYM																																							
CMD >S																			1	0	1	2	3	4	5	6	7												
012A 89F3 MOV BX,SI																			DS:0000	CD	20	FF	9F	00	EA	F0	FE												
012C B8004C MOV AX,4C00																			DS:0008	AD	DE	1B	05	C5	06	00	00												
012F CD21 INT 21																			DS:0010	18	01	10	01	18	01	92	01												
0131 8B5702 MOV DX,[BX+02]																			DS:0018	01	01	01	00	02	FF	FF	FF												
0134 85D2 TEST DX,DX																			DS:0020	FF	FF	FF	FF	FF	FF	FF	FF												
0136 7504 JNZ 013C																			DS:0028	FF	FF	FF	FF	EB	19	C0	11												
0138 85C0 TEST AX,AX																			DS:0030	A2	01	14	00	18	00	F5	19												
013A 741C JZ 0158																			DS:0038	FF	FF	FF	FF	00	00	00	00												
013C C746DC0000 MOV [BP-24],0000																			DS:0040	05	00	00	00	00	00	00	00												
																			DS:0048	00	00	00	00	00	00	00	00	00											
2	3	4	5	6	7	8	9	A	B	C	D	E	F	0	1	2																							
DS:0103	E8	03	E4	07	DC	05	94	11	1A	04	BB	00	00	B8	94	11	8.S.0. .																						

Task 3: Assume that there is a class of five people with the following grades: 700, 870, 916, 550, and 313. Find the highest grade.

- Use Conditional Jumps
- Use Compare statement for both termination of loop and compare numbers.
- Use of CX is Prohibited
- Find highest grade and write memory location of grade and value.

org 0x100

jmp start

class: dw 700h, 870h, 916h, 550h, 313h

start:

mov bx, 0

mov ax, [class]

mov si, 2

jmp l1

greater:

mov ax, [class + si]

mov bx, 0

mov bx, si

add bx, class ; store the highest grade's address in BX,

add si, 2

jmp l1

l1:

cmp [class + si], ax

jg greater

add si, 2

cmp si, 10

jnz l1

mov ax, 0x4c00

int 0x21

The highest grade is at: 0107

AX	0916	SI	000A	CS	19F5	IP	013D	Stack	+0 0000	Flags	7244
BX	0107	DI	0000	DS	19F5				+2 20CD		
CX	0042	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF	DF IF SF ZF AF PF CF
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0	0 1 0 1 0 1 0

CMD >						1 0 1 2 3 4 5 6 7															
						DS:0000 CD 20 FF 9F 00 EA F0 FE															
013B 75F0 JNZ 012D						DS:0008 AD DE 1B 05 C5 06 00 00															
013D B8004C MOV AX,4C00						DS:0010 18 01 10 01 18 01 92 01															
0140 CD21 INT 21						DS:0018 01 01 01 00 02 FF FF FF															
0142 5E POP SI						DS:0020 FF FF FF FF FF FF FF FF															
0143 FC CLD						DS:0028 FF FF FF FF EB 19 C0 11															
0144 837D0E00 CMP [DI+0E],0000						DS:0030 A2 01 14 00 18 00 F5 19															
0148 7409 JZ 0153						DS:0038 FF FF FF FF 00 00 00 00															
014A 8B46F2 MOV AX,[BP-0E]						DS:0040 05 00 00 00 00 00 00 00															
014D 48 DEC AX						DS:0048 00 00 00 00 00 00 00 00															

2		3 4 5 6 7 8 9 A B C D E F 0 1 2																															
DS:0103		00 07 70 08 16 09 50 05 13 03 BB 00 00 A1 03 01																...p...P. ...η...ί..															
DS:0113		BE 02 00 E9 14 00 8B 84 03 01 BB 00 00 89 F3 81																J...θ...Yā ...η...ë&u															
DS:0123		C3 03 01 81 C6 02 00 E9 00 00 39 84 03 01 7F E6																J...üþ...θ ...9ā...øµ															
DS:0133		81 C6 02 00 81 FE 0A 00 75 F0 B8 00 4C CD 21 5E																üþ...üµ... u≡γ.L=!?^															
DS:0143		FC 83 7D 0E 00 74 09 8B 46 F2 48 3B 46 F6 7E 08																"āþ...t.t.î F2H;F+~															